

Void Analysis Report

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1. Overview

This document summarizes the complete workflow, datasets, methods, and results associated with the ongoing void environment Hilbert-Huang Transform (HHT) analysis of supernova (SN) residuals. The investigation aims to identify whether variations in SN residual energy (E_{IMF2}) show any correlation ("locking") with Baryon Acoustic Oscillation (BAO) scales, and whether such effects vary across large-scale structure environments (field, rim, wall, interior).

2. Datasets and Sources

Includes SN residuals, redshift, environmental classifications, weighting, and BAO datasets (DR12 void-matched). Describes dataset suitability and structure.

3. Scripts and Functions

Summarizes `voids_hht.py`, `lock_env_advanced.py`, and supporting utilities, with parameter details and output file structures.

4. Results Summary

Provides correlation results (r , p -values) by environment, IMF2 per z -bin table, and interpretation of non-locking outcomes.

5. Diagnostic Conclusions

No significant IMF2–BAO locking detected. Field environment dominates statistics; rim/wall subsets insufficient. IMF2 peaks likely artifacts.

6. Recommendations for Future Work

Suggests merged environments, fewer bins, winsorization, and alternative regression testing for phase coherence.

7. Rebuilding the Analysis Environment

Details Python environment setup, dependencies, directory structure, and command-line execution sequences.

8. Summary

DR12-based void-environment HHT analysis shows no measurable IMF2–BAO phase coupling, but framework remains valid for future high-density datasets.